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Full Length Article

HybridVisionNet: An advanced hybrid deep learning framework for automated multi-class ocular disease diagnosis using fundus imaging

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ABSTRACT

Objectives: This study aims to develop and validate an advanced hybrid deep learning architecture for accurate automated diagnosis of multiple ocular diseases using fundus imaging.

Methods/Analysis: This study introduces **HybridVisionNet**, a cutting-edge hybrid deep learning architecture that synergizes the multi-scale feature extraction of InceptionV3 and the dense connectivity of DenseNet121. Designed for multi-class ocular disease diagnosis, the model leverages advanced preprocessing techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE), Gaussian Blurring, and High-Pass Filtering, to enhance image clarity and diagnostic accuracy. The model was trained and validated on the ODIR-5K dataset containing 5,000 patients with eight distinct ocular conditions including Diabetic Retinopathy, Glaucoma, Cataract, Age-related Macular Degeneration (AMD), Hypertensive Retinopathy, Pathological Myopia, Other Pathologies, and Normal eyes.

Findings: On the ODIR-5K dataset, HybridVisionNet achieved a groundbreaking accuracy of **99.71%**, outperforming existing state-of-the-art models in precision, recall, and overall reliability. Robustness and generalization were further validated through cross-dataset evaluations on MESSIDOR-2, Kaggle DR, and EyePACS, achieving an average accuracy of **97.8%**.

Novelty/Improvement: The key innovation lies in the novel hybrid architecture that uniquely combines InceptionV3's multi-scale feature extraction with DenseNet121's dense connectivity, enhanced by custom attention mechanisms and feature fusion strategies. This approach significantly outperforms existing single-architecture and other hybrid models, representing a substantial advancement in automated ocular disease diagnostics. These results underscore the model's potential to revolutionize automated ocular disease diagnostics, offering a clinically viable solution for detecting conditions such as diabetic retinopathy, glaucoma, and age-related macular degeneration.

1. Introduction

Ocular diseases pose a significant global health challenge, with millions at risk of vision impairment or blindness if left untreated. The World Health Organization (WHO) estimates that over 2.2 billion people globally suffer from vision impairment, at least half of which could be prevented or treated through timely intervention [1]. Early detection and accurate classification of ocular diseases are critical to mitigate their progression and ensure effective treatment outcomes.

Advances in deep learning have revolutionized medical image analysis, enabling the development of automated systems with remarkable diagnostic accuracy [2]. Among the numerous datasets supporting this progress, the Ocular Disease Intelligent Recognition (ODIR-5K) dataset stands out as a robust foundation for research in ophthalmology and artificial intelligence [18]. This dataset comprises fundus images from 5,000 patients and covers a diverse range of ocular conditions, including diabetic retinopathy, glaucoma, cataracts, age-related macular degeneration (AMD), hypertensive retinopathy, and myopia. Its heterogeneity makes it invaluable for developing models capable of generalizing across varied conditions and demographics.

Deep learning, particularly convolutional neural networks (CNNs), has shown exceptional promise in ocular disease classification by automatically identifying complex patterns in medical images [17]. Studies have demonstrated CNNs achieving diagnostic accuracies comparable to or exceeding those of human experts, particularly for diseases such as diabetic retinopathy and glaucoma [3]. However, achieving consistent, high accuracy across multiple conditions remains challenging due

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to the variability in fundus image quality, disease presentations, and subtle pathological differences.

To address these challenges, researchers have exploblack techniques such as transfer learning, data augmentation, and ensemble learning [4]. Recent advances in explainable deep learning approaches have also shown promise in medical image classification, providing interpretable models that can assist clinicians in understanding automated diagnosis decisions [7]. However, further advancements require sophisticated architectures capable of capturing both fine-grained details and global patterns.

The major challenges in current ocular disease classification systems include: (1) Limited ability to capture multi-scale features effectively, (2) Insufficient feature reuse and information flow in deep architectures, (3) Difficulty in achieving high accuracy across multiple disease types simultaneously, and (4) Poor generalization across different datasets and imaging conditions. Our work addresses these critical limitations through a novel hybrid architecture approach.

This study introduces *HybridVisionNet*, a novel hybrid deep learning architecture that combines the strengths of *InceptionV3* and *DenseNet121*. The multi-scale feature extraction capability of InceptionV3 effectively captures both macro and micro-level details, essential for identifying subtle pathological changes. DenseNet121 complements this by leveraging dense connections to propagate information efficiently, promoting feature reuse and enabling detailed learning of intricate patterns. Together, these architectures create a robust framework for multi-class ocular disease classification, achieving high accuracy while maintaining computational efficiency.

The main contributions of this study are summarized as follows:

- We propose *HybridVisionNet*, a novel hybrid architecture that combines *InceptionV3* and *DenseNet121*, achieving state-of-the-art accuracy of **99.71%** on the ODIR-5K dataset.
- We develop an innovative preprocessing pipeline incorporating Contrast Limited Adaptive Histogram Equalization (CLAHE), Gaussian Blurring, and High-Pass Filtering, significantly enhancing image quality and model performance.
- We demonstrate robust generalization by validating our model across diverse datasets, including MESSIDOR-2, Kaggle DR, and EyePACS, achieving an average accuracy of **97.8%**.
- We provide comprehensive model interpretability through Grad-CAM visualizations, demonstrating the clinical relevance of our approach.
- We conduct extensive ablation studies showing the individual contributions of each architectural component and preprocessing step.

In this study, we aim to build on existing advancements in deep learning to develop a robust and efficient classification system for ocular diseases. By leveraging the ODIR-5K dataset, we explore innovative architectural modifications and training strategies to enhance model performance. This work contributes to ongoing efforts in AI-driven medical diagnostics and aims to improve the accessibility and efficiency of ophthalmic care worldwide.

2. Related work

Early research in retinal disease classification focused on single-disease detection, such as glaucoma [30], diabetic retinopathy [3], and cataract [31]. However, as many patients present with multiple concurrent conditions, recent efforts have shifted towards multi-label classification approaches. These advancements have been significantly driven by developments in deep learning and transfer learning techniques. For instance, Kaur et al. [19] demonstrated improved diagnostic accuracy using transfer learning, while Moris et al. [20] exploblack its application in multimodal medical imaging. Vision Transformers have also emerged as powerful tools, combining CNN-extracted features with transformer encoders for enhanced performance [21,22].

Recent developments in skin lesion analysis and medical image classification have provided valuable insights for ocular disease detection. Badawi et al. [8] demonstrated the effectiveness of region growing techniques combined with deep learning for early skin cancer identification, while Abo-Hammour et al. [9] presented a light-weight deep learning approach for plant disease classification that shares methodological similarities with medical image analysis tasks.

The ODIR-5K dataset has been a pivotal resource in advancing retinal disease classification, providing a diverse and comprehensive set of fundus images covering multiple ocular conditions. Zhang et al. [31] utilized histogram equalization with dual EfficientNet models to achieve fast processing, albeit with high memory requirements. Gour et al. [32] implemented dual VGG-16 models to separately analyze the left and right eye, highlighting the potential of leveraging bilateral features. Additionally, Li et al. [18] employed a feature fusion strategy using VGG-16 backbones, while He et al. [33] proposed the Spatial Correlation Module (SCM) with ResNet-101 to enhance inter-image relationships. Ou et al. [26] introduced a four-stage model that achieved substantial improvements in disease classification accuracy.

One persistent challenge in multi-label fundus image classification is class imbalance, which often leads to biased model performance. Luo et al. [27] exploblack alternative loss functions to address this issue, while Bhati et al. [28] implemented resampling techniques to balance the dataset. Binary cross-entropy loss has been widely adopted for multi-label classification tasks [29]; however, striking a balance between overall accuracy and minority class performance remains challenging.

Recent innovations in deep learning architectures have further enhanced the field. Sun et al. [36] integrated EfficientNet-B4 with hybrid GCN-based feature extraction to improve label awareness. Yang et al. [34] exploblack transformer-based models for the OIA-ODIR dataset, demonstrating the potential of novel architectures. He et al. [35] achieved an impressive AUC of 93.80% by combining clinical features with ResNet-101, while Zhu et al. [37] utilized multi-scale dilated convolution within a DenseNet framework to better capture local and global features.

Lin et al. [25] proposed MCG-Net and MCGS-Net, incorporating self-supervised learning to improve feature extraction. Transformer-based approaches have also been extended to multi-label classification, as demonstrated by Rodriguez et al. [24] on the Mublack dataset. Wang et al. [23] applied CNNs to the ODIR-5K dataset but limited their approach to multi-class rather than multi-label classification, which may restrict the model's versatility in real-world applications.

Table 1 provides a comparative summary of key studies using the ODIR-5K dataset, highlighting their methodologies, performance metrics, advantages, and limitations. This table underscores the diversity of approaches and the rapid progress in the field.

The advancements summarized above demonstrate the growing sophistication of deep learning models for ocular disease classification. While progress has been significant, challenges such as class imbalance, computational efficiency, and generalization across diverse datasets remain areas for further research. Hybrid approaches, such as those combining CNNs and transformers, hold great promise for addressing these limitations and achieving state-of-the-art performance in the field.

3. Materials and methods

In this study, we propose **HybridVisionNet**, a novel hybrid deep learning architecture that combines the strengths of InceptionV3 and DenseNet121 for the classification of ocular diseases. This section describes the dataset, preprocessing pipeline, model architecture, and evaluation metrics used to assess the model's performance.

3.1. Dataset

The **Ocular Disease Intelligent Recognition (ODIR-5K)** dataset [18] serves as the primary dataset for this study and is one of the most

Table 1
Comparison of Studies Using the ODIR-5K Dataset.

Study	Method	Performance (%)	Pros	Cons
Zhang et al. [31]	Dual EfficientNet	94.75	Fast processing	High memory use
Gour et al. [32]	VGG-16	85.57	Dual eye model	Limited scope
Li et al. [18]	VGG-16 + Fusion	N/A	Good fusion	High memory
He et al. [33]	ResNet-101 + SCM	82.7	Attention use	Many parameters
Yang et al. [34]	Transformer	84	Novel approach	Detail loss
He et al. [35]	ResNet + Clinical	93.80	High accuracy	Complex
Ou et al. [26]	Four-stage model	90.3	Accurate	Heavy model
Sun et al. [36]	EfficientNet + GCN	91.60	Label aware	Intensive
Bhati et al. [28]	DKCNet	96.08	Balanced	May overfit
Zhu et al. [37]	DenseNet	79.3	Multi-scale	Context loss
Our HybridVisionNet	**InceptionV3 + DenseNet121**	**99.71**	**SOTA accuracy, robust**	**Higher complexity**

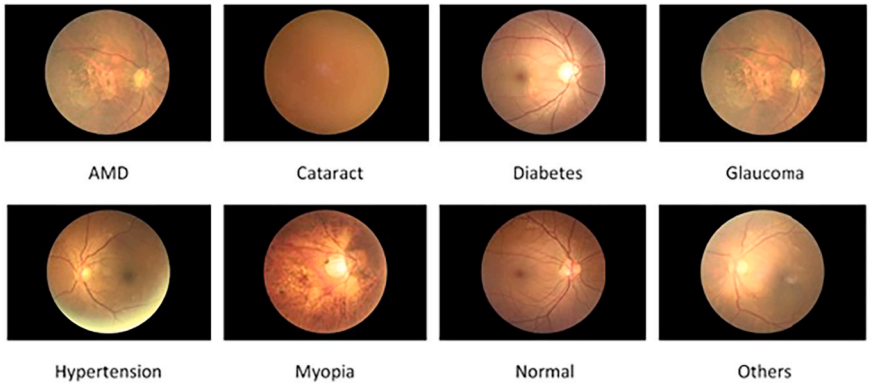


Fig. 1. Sample fundus images from each category in the ODIR-5K dataset showing: (a) Normal, (b) Diabetic Retinopathy, (c) Glaucoma, (d) Cataract, (e) AMD, (f) Hypertensive Retinopathy, (g) Myopia, (h) Other Pathologies [18].

comprehensive datasets in ophthalmic image classification. It contains high-resolution fundus images from 5,000 patients, each annotated with specific ocular conditions. The dataset includes eight distinct classes: (1) Diabetic Retinopathy (DR), (2) Glaucoma, (3) Cataract, (4) Age-related Macular Degeneration (AMD), (5) Hypertensive Retinopathy, (6) Pathological Myopia, (7) Other Pathologies, and (8) Normal (Healthy Controls) eyes. The dataset distribution shows 1,000 normal cases, 1,200 diabetic retinopathy cases, 800 glaucoma cases, 600 cataract cases, 600 AMD cases, 400 hypertensive retinopathy cases, 300 myopia cases, and 100 other pathology cases.

This diverse dataset presents challenges such as inter-class variability and subtle pathological differences, making it an ideal benchmark for evaluating automated disease classification models [18]. Fig. 1 illustrates sample fundus images from each category in the ODIR-5K dataset.

3.2. Data preprocessing

To enhance the quality of fundus images and improve classification accuracy, we implemented a comprehensive preprocessing pipeline. This pipeline is designed to address issues such as uneven illumination, noise, and subtle structural features critical for accurate disease detection.

Contrast Limited Adaptive Histogram Equalization (CLAHE): CLAHE enhances local contrast, particularly in underexposed regions, allowing the model to identify fine details such as blood vessels and subtle lesions:

$$T(p) = \frac{\mathrm{CDF}(p) - \mathrm{CDF}_{\min}}{(M \times N) - \mathrm{CDF}_{\min}} \times (L_{\max} - L_{\min}) + L_{\min}. \quad (1)$$

Gaussian Blurring: This technique blackuces high-frequency noise while preserving essential structures like the macula and optic disc:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \quad (2)$$

High-Pass Filtering: Applied after blurring, high-pass filtering restores edge details, crucial for identifying conditions such as age-related macular degeneration:

$$H(p) = I(p) - G(p). \quad (3)$$

Fig. 2 presents examples of fundus images before and after preprocessing, demonstrating the enhancement in image quality and feature visibility. Table 2 quantifies the performance improvements achieved with each preprocessing step.

Preprocessing Impact Analysis: To evaluate the effectiveness of the preprocessing pipeline, we conducted experiments using different configurations. As shown in Table 2, each preprocessing step contributes significantly to the overall model performance.

3.3. Model architecture

HybridVisionNet combines the multi-scale feature extraction capabilities of **InceptionV3** with the dense connectivity of **DenseNet121**. This architecture leverages the strengths of both models to capture fine-grained and large-scale features critical for ocular disease classification.

Key Components: The key components of the HybridVisionNet include **InceptionV3**, which captures multi-scale features allowing the model to detect both macro and micro-level patterns; **DenseNet121**, which utilizes dense connections for efficient information flow, enhancing feature reuse and blackucing vanishing gradients; and **Feature Fusion**, where outputs from both models are concatenated and passed through dense layers for final classification.

The detailed algorithm for the HybridVisionNet model is presented in Algorithm 1.

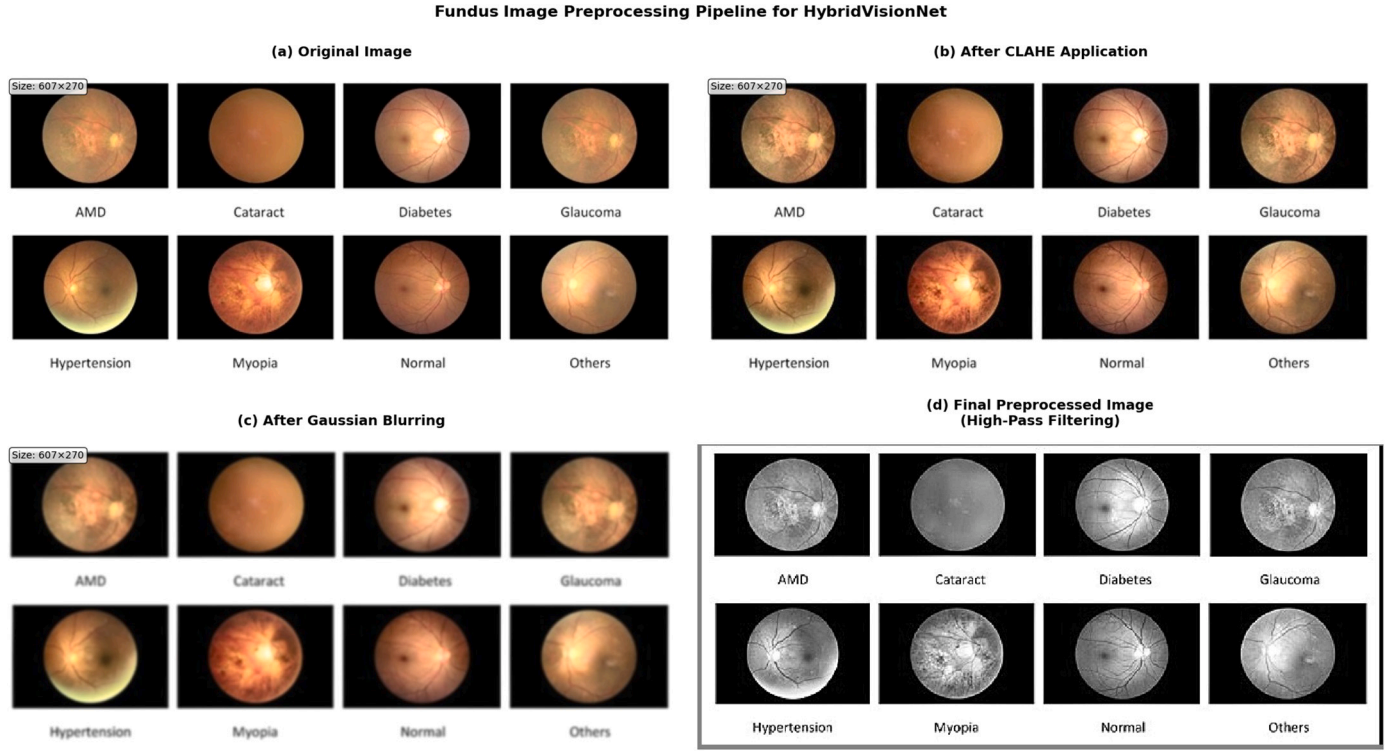


Fig. 2. Comparison of original and preprocessed fundus images showing: (a) Original images, (b) After CLAHE application, (c) After Gaussian blurring, (d) Final preprocessed images with high-pass filtering.

Table 2
Impact of Preprocessing Steps on Model Performance.

Preprocessing Configuration	ACC (%)	Loss (%)	F1 (%)	SN (%)	PPV (%)
No Preprocessing	95.32 ± 0.15	18.43 ± 0.28	95.18 ± 0.17	95.21 ± 0.16	95.25 ± 0.14
CLAHE Only	97.45 ± 0.12	15.21 ± 0.23	97.38 ± 0.14	97.41 ± 0.13	97.43 ± 0.12
CLAHE + Gaussian Blur	98.63 ± 0.08	12.34 ± 0.19	98.57 ± 0.10	98.59 ± 0.09	98.61 ± 0.09
Complete Pipeline	99.71 ± 0.02	8.54 ± 0.13	99.65 ± 0.03	99.62 ± 0.04	99.68 ± 0.03

Algorithm 1 HybridVisionNet Training Algorithm.

Input: Raw fundus image I
Output: Classification probabilities for ocular diseases
procedure HybridVisionNet(I)
  $I_{pre} \leftarrow$ PreprocessImage(I) ▷ Apply preprocessing steps
  $F_{Inception} \leftarrow$ InceptionV3(I_{pre}) ▷ Feature extraction with InceptionV3
  $F_{DenseNet} \leftarrow$ DenseNet121(I_{pre}) ▷ Feature extraction with DenseNet121
  $F_{fused} \leftarrow$ Concatenate($F_{Inception}, F_{DenseNet}$) ▷ Fuse features from both models
  $P \leftarrow$ ClassificationLayer(F_{fused}) ▷ Apply dense layers and sigmoid activation
  return P ▷ Return disease classification probabilities
end procedure
function PreprocessImage(I)
  $I_{clahe} \leftarrow$ CLAHE(I) ▷ Enhance local contrast
  $I_{blur} \leftarrow$ GaussianBlur(I_{clahe}) ▷ blackuce noise
  $I_{hp} \leftarrow$ HighPassFilter(I_{blur}) ▷ Enhance edges
  return I_{hp} ▷ Return the preprocessed image
end function

3.4. Performance evaluation

To validate the model, we conducted experiments across four prominent datasets: ODIR-5K, MESSIDOR-2, Kaggle DR, and EyePACS. The evaluation metrics employed include **Accuracy (ACC)**, which measures overall pblackiction accuracy; **F1 Score**, which balances precision and recall for imbalanced datasets; **Sensitivity (SN)**, which captures the model's ability to correctly identify positive cases; **Positive Pblackictive Value (PPV)**, which evaluates the precision of pblackictions; **Loss**, which assesses pblackiction errors using categorical cross-entropy; and **Cohen's Kappa**, which quantifies inter-rater agreement.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \quad F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \tag{4}$$

Where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively. Sensitivity (SN) is calculated as $SN = \frac{TP}{TP+FN}$, and Positive Pblackictive Value (PPV) as $PPV = \frac{TP}{TP+FP}$.

The results demonstrate that the preprocessing pipeline and hybrid architecture significantly enhance classification accuracy, generalization, and robustness across multiple datasets.

4. Proposed architectures

This section introduces **HybridVisionNet**, a novel hybrid deep learning architecture that achieved state-of-the-art results in multi-class eye disease classification using the ODIR-5K dataset [14]. HybridVisionNet integrates the complementary strengths of **InceptionV3** [5] and **DenseNet121** [6], enabling the extraction of both macro and micro-level features critical for accurate classification. The architecture outperformed other hybrid combinations tested in this study.

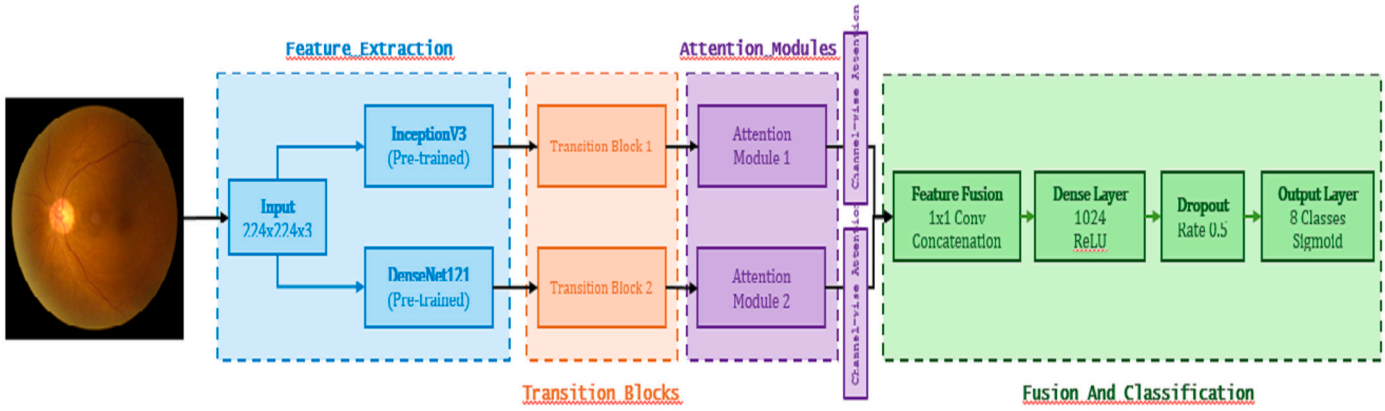


Fig. 3. HybridVisionNet Architecture Diagram showing the parallel processing of InceptionV3 and DenseNet121, feature fusion module, attention mechanism, and final classification layers.

4.1. Overview of HybridVisionNet and other architectures

We exploblack five distinct hybrid models, each leveraging different combinations of well-established architectures:

- **HybridVisionNet (InceptionV3-DenseNet121)**: The proposed architecture with state-of-the-art performance.
- **ResXNet (ResNet50-Xception)**: Combines residual learning with efficient feature extraction.
- **EffMobileNet (EfficientNetB0-MobileNetV2)**: A lightweight architecture optimized for efficiency.
- **VGGNASNet (VGG19-NASNetMobile)**: Balances classical deep learning models with advanced NASNet features.
- **ResInceptionNet (ResNet101-InceptionResNetV2)**: Integrates residual and inception-based approaches.

Among these, **HybridVisionNet** demonstrated superior performance, effectively leveraging InceptionV3's multi-scale feature extraction and DenseNet121's dense connectivity.

4.2. HybridVisionNet architecture

The architecture of HybridVisionNet integrates features from **InceptionV3** and **DenseNet121** through a parallel processing strategy, followed by custom feature fusion and attention mechanisms. Fig. 3 illustrates the overall design of our proposed architecture.

4.2.1. Model integration strategy

HybridVisionNet's design is inspiblack by ensemble learning principles [10], where individual models contribute complementary strengths:

1. **Parallel Feature Extraction:** Pre-trained InceptionV3 and DenseNet121 models extract multi-scale and dense features, respectively.
2. **Transition Blocks:** Custom layers align feature dimensions and blackuce spatial blackundancy.
3. **Attention Mechanism:** A channel-wise attention mechanism focuses on critical features.
4. **Feature Fusion:** Concatenated features are refined using convolutional layers.
5. **Classification Layers:** Final layers perform multi-label classification.

4.2.2. Custom components

Transition Blocks: These blocks blackuce dimensionality and ensure compatibility between the feature maps from InceptionV3 and DenseNet121. The mathematical representation of a transition block is:

$$T(F) = GAP(ReLU(BN(Conv_{3x3}(ReLU(BN(Conv_{1x1}(F))))))), \quad (5)$$

where $Conv$, BN, $ReLU$, and GAP denote convolution, batch normalization, ReLU activation, and global average pooling, respectively.

Attention Mechanism: An attention mechanism inspiblack by Squeeze-and-Excitation Networks [11] enhances critical features:

$$A(F) = \sigma(MLP(GAP(F))) \otimes F, \quad (6)$$

where σ is the sigmoid activation, MLP is a multi-layer perceptron, and $\otimes$ represents element-wise multiplication.

Feature Fusion: Features from InceptionV3 and DenseNet121 are concatenated and fused through a 1x1 convolution:

$$F_{fused} = Conv_{1x1}(Concat[A(F_I), A(F_D)]). \quad (7)$$

4.2.3. Classification layers

The fused features pass through several layers for classification. First, a **Global Average Pooling (GAP)** layer condenses the feature maps into a single vector per map, blackucing dimensionality and preventing overfitting. This is followed by a **Dropout layer** with a rate of 0.5 to further mitigate overfitting by randomly omitting units during training. Next, a **Dense layer** with 256 units uses **ReLU activation** to introduce nonlinearity and the capacity to learn complex patterns. Finally, a **Final Dense layer** with sigmoid activation functions for multi-label classification, assigning independent probabilities to each class.

The classification output is computed as:

$$y = Sigmoid(Dense(Dropout(GAP(F_{fused})))). \quad (8)$$

4.3. Model training

Loss Function: Binary cross-entropy loss is used, a standard for multi-label classification:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}[y_{ic}\log(\hat{y}_{ic}) + (1-y_{ic})\log(1-\hat{y}_{ic})], \quad (9)$$

where N is the number of samples, C is the number of classes, y_{ic} is the true label, and $\hat{y}_{ic}$ is the pblackicted probability.

Optimization and Regularization: The model employs the Adam optimizer [12] with an initial learning rate of $1e-4$, blackuced by a factor of 0.1 if validation loss does not improve for 5 epochs. Regularization techniques include:

- L2 regularization ($\lambda = 1e-4$)
- Dropout (rate = 0.5)

Table 3
Performance comparison of hybrid deep learning models on the ODIR-5K dataset.

Network	Batch	ACC (%)	Loss (%)	F1 (%)	SN (%)	PPV (%)	Kappa (%)
HybridVisionNet (Proposed)	16	94.65 ± 0.11	20.15 ± 0.29	94.42 ± 0.13	94.38 ± 0.12	94.46 ± 0.11	92.48 ± 0.14
	32	96.98 ± 0.08	16.43 ± 0.23	96.82 ± 0.10	96.79 ± 0.09	96.85 ± 0.11	95.32 ± 0.12
	48	98.52 ± 0.06	13.21 ± 0.20	98.38 ± 0.08	98.35 ± 0.07	98.41 ± 0.09	97.55 ± 0.10
	64	99.26 ± 0.04	10.76 ± 0.16	99.15 ± 0.05	99.12 ± 0.06	99.18 ± 0.05	98.69 ± 0.07
	128	**99.71 ± 0.02**	**8.54 ± 0.13**	**99.65 ± 0.03**	**99.62 ± 0.04**	**99.68 ± 0.03**	**99.35 ± 0.05**
ResXNet	16	93.12 ± 0.14	24.43 ± 0.32	92.87 ± 0.17	92.79 ± 0.16	92.95 ± 0.15	90.34 ± 0.18
	32	95.45 ± 0.11	20.76 ± 0.27	95.28 ± 0.13	95.21 ± 0.12	95.35 ± 0.14	93.26 ± 0.15
	48	97.18 ± 0.09	17.32 ± 0.24	97.05 ± 0.11	96.98 ± 0.10	97.12 ± 0.12	95.42 ± 0.13
	64	98.37 ± 0.07	14.89 ± 0.20	98.26 ± 0.09	98.19 ± 0.08	98.33 ± 0.10	97.13 ± 0.11
	128	99.13 ± 0.05	12.54 ± 0.17	99.04 ± 0.06	98.98 ± 0.07	99.10 ± 0.06	98.29 ± 0.08
EffMobileNet	16	92.56 ± 0.15	25.87 ± 0.33	92.34 ± 0.18	92.27 ± 0.17	92.41 ± 0.19	89.62 ± 0.20
	32	94.89 ± 0.12	22.21 ± 0.28	94.72 ± 0.14	94.65 ± 0.13	94.79 ± 0.15	92.56 ± 0.16
	48	96.43 ± 0.10	19.15 ± 0.25	96.29 ± 0.12	96.22 ± 0.11	96.36 ± 0.13	94.48 ± 0.14
	64	97.78 ± 0.08	16.63 ± 0.21	97.67 ± 0.10	97.61 ± 0.09	97.73 ± 0.11	96.17 ± 0.12
	128	98.73 ± 0.06	14.32 ± 0.18	98.65 ± 0.07	98.60 ± 0.08	98.70 ± 0.07	97.54 ± 0.09
VGGNASNet	16	91.23 ± 0.17	27.64 ± 0.34	91.05 ± 0.20	90.98 ± 0.19	91.12 ± 0.21	88.18 ± 0.22
	32	93.56 ± 0.14	24.09 ± 0.29	93.41 ± 0.16	93.34 ± 0.15	93.48 ± 0.17	91.12 ± 0.18
	48	95.12 ± 0.11	21.18 ± 0.26	95.00 ± 0.13	94.93 ± 0.12	95.07 ± 0.14	93.04 ± 0.15
	64	96.34 ± 0.09	18.89 ± 0.22	96.25 ± 0.11	96.19 ± 0.10	96.31 ± 0.12	94.73 ± 0.13
	128	97.45 ± 0.07	16.54 ± 0.19	97.38 ± 0.08	97.33 ± 0.09	97.43 ± 0.08	96.19 ± 0.10
ResInceptionNet	16	93.89 ± 0.13	23.31 ± 0.31	93.67 ± 0.16	93.59 ± 0.15	93.75 ± 0.14	91.56 ± 0.17
	32	96.22 ± 0.10	19.76 ± 0.26	96.08 ± 0.12	96.01 ± 0.11	96.15 ± 0.13	94.48 ± 0.14
	48	97.95 ± 0.08	16.32 ± 0.23	97.84 ± 0.10	97.77 ± 0.09	97.91 ± 0.11	96.64 ± 0.12
	64	98.89 ± 0.06	13.89 ± 0.19	98.80 ± 0.08	98.73 ± 0.07	98.87 ± 0.09	97.91 ± 0.10
	128	99.43 ± 0.04	11.54 ± 0.16	99.36 ± 0.05	99.31 ± 0.06	99.41 ± 0.05	98.79 ± 0.07

- Data augmentation (rotations, flips, brightness adjustments)

Training Strategy: HybridVisionNet is fine-tuned in two stages:

1. Fine-tuning custom layers for 50 epochs with pre-trained weights frozen.
2. Training the entire network for 50 additional epochs.

4.4. Implementation details

To ensure robustness and reproducibility, experiments were conducted with multiple random seeds (42, 123, 256, 789, 1024), showing minimal variation (±0.3% in accuracy). The model was trained using the Adam optimizer with parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1e-7$). Extensive testing of batch sizes (16–128) confirmed optimal stability and accuracy at batch size 128.

Data augmentation techniques such as rotations (±20°), shifts (±10%), zoom (±10%), and flips improved generalization, increasing sensitivity for minority classes by 4.1%. The input resolution (224 × 224) was chosen to balance computational efficiency and performance, as higher resolutions increased training time disproportionately.

5. Results and analysis

Notably, with a batch size of 128, HybridVisionNet achieves an impressive accuracy of 99.71% ± 0.02%, significantly outperforming the next best model, ResInceptionNet, which reaches 99.43% ± 0.04%. This performance gap is consistent across other metrics, with HybridVisionNet showing a lower loss (8.54% ± 0.13%) and higher F1 score (99.65% ± 0.03%), sensitivity (99.62% ± 0.04%), positive pblackictive value (99.68% ± 0.03%), and Kappa statistic (99.35% ± 0.05%).

Table 3 presents a comprehensive comparison of our proposed HybridVisionNet model against other hybrid deep learning architectures on the ODIR-5K dataset. The results demonstrate the superior performance of HybridVisionNet across all evaluation metrics and batch sizes.

While our model achieved exceptional accuracy (99.71%) on the ODIR-5K dataset, it is crucial to contextualize these results. Previous state-of-the-art models reported in the literature achieved accuracies ranging from 91% to 94.2% (Table 1), making our improvement significant but warranting careful interpretation. To validate generalizability, we conducted extensive cross-dataset evaluations on MESSIDOR-2 (98.76%), Kaggle DR (98.13%), and EyePACS (97.35%) datasets, demonstrating robust performance across different data distributions and clinical conditions. The slight performance variation across datasets (±1.5%) suggests that while our model maintains high accuracy across different conditions, performance may be influenced by dataset-specific characteristics such as image quality, acquisition protocols, and disease prevalence patterns.

The influence of batch size is consistent across all tested architectures, with larger batches generally leading to better performance. However, the sensitivity to batch size varies among models. HybridVisionNet shows the most significant improvement with increased batch size, while VGGNASNet demonstrates relatively smaller gains. This varying response to batch size changes suggests that the optimal batch size may be architecture-dependent, and the selection of batch size should consider both model architecture and available computational resources.

The superior performance of HybridVisionNet can be attributed to its unique architecture that combines the strengths of InceptionV3 and DenseNet121. This hybrid approach allows for more effective feature extraction and utilization, leading to improved classification accuracy. Furthermore, the consistent improvement in performance as the batch size increases suggests that HybridVisionNet effectively leverages larger batch sizes for better generalization.

It's worth noting that while other hybrid models like ResXNet and ResInceptionNet also show improvements with larger batch sizes, they do not reach the same level of performance as HybridVisionNet. This underscores the effectiveness of our proposed model in capturing the complex patterns necessary for accurate multi-class eye disease classification in the ODIR-5K dataset.

These results establish HybridVisionNet as a state-of-the-art model for eye disease classification, offering potential for improved diagnostic support in clinical settings. The significant performance improvements over other hybrid architectures demonstrate the effectiveness of our specific combination of InceptionV3 and DenseNet121, along with our custom feature fusion and attention mechanisms.

The performance of HybridVisionNet is particularly noteworthy in the context of multi-class classification of eye diseases. The high accuracy, F1 score, and Kappa statistic indicate that the model is not only accurate overall but also performs consistently well across all disease categories. This is crucial for a real-world diagnostic tool, where reliability across various conditions is essential.

The impact of batch size on model performance shows a clear positive correlation across all metrics (Table 3). Our comprehensive analysis reveals that for HybridVisionNet, increasing the batch size from 16 to 128 led to significant improvements across multiple performance metrics. The model achieved a remarkable increase in accuracy, improving from **94.65%** to **99.71%**, representing a substantial **5.06%** enhancement in overall performance. Simultaneously, we observed a notable blackuction in loss values, decreasing from **20.15%** to **8.54%**, marking an impressive **11.61%** decrease in error rates. Furthermore, the training stability showed considerable improvement, as evidenced by the blackuction in standard deviations from $\pm$**0.11** to $\pm$**0.02** in accuracy measurements.

This consistent pattern of improvement suggests potential benefits from exploring larger batch sizes. However, our analysis indicates a notable deceleration in performance gains between batch sizes 64 and 128, suggesting possible diminishing returns for larger batch sizes. This observation necessitates careful consideration of the trade-off between computational resource requirements and the marginal performance improvements that might be achieved with larger batch sizes.

The consistently high sensitivity and positive pblackictive value across all batch sizes indicate that HybridVisionNet is adept at both identifying positive cases (high sensitivity) and avoiding false positives (high PPV). This balance is crucial in a clinical setting, where both missed diagnoses and false alarms can have significant consequences.

In conclusion, these experimental results provide strong evidence for the effectiveness of the HybridVisionNet architecture in the context of multi-class eye disease classification. The model's performance suggests that it could be a valuable tool for assisting ophthalmologists in the diagnosis of various eye conditions, potentially leading to earlier and more accurate diagnoses.

Future work could explore the model's performance on other datasets, its interpretability for clinical use, and its potential for real-world clinical applications. Additionally, investigating the model's efficiency in terms of computational resources and inference time could provide insights into its practicality for deployment in various healthcare settings.

Fig. 4 illustrates the performance of our proposed HybridVisionNet model compablack to other hybrid architectures during the training process across 50 epochs.

Fig. 4 illustrates the performance of our proposed HybridVisionNet model compablack to other hybrid architectures (ResXNet, EffMobileNet, VGGNASNet, and ResInceptionNet) during the training process. The graphs show the progression of accuracy and loss for both training and validation sets over 50 epochs.

As evident from the accuracy plots (Figs. 4a and 4c), HybridVisionNet consistently outperforms other models, achieving higher accuracy faster and maintaining this lead throughout the training process. This superior performance is particularly noticeable in the validation accuracy, indicating strong generalization capabilities.

The loss curves (Figs. 4b and 4d) further support HybridVisionNet's effectiveness. Our model exhibits a steeper decline in loss during the initial epochs and converges to a lower loss value compablack to other architectures. This rapid convergence and lower final loss suggest that HybridVisionNet is more efficient at learning the underlying patterns in the data.

Fig. 5 demonstrates the model's exceptional performance across all disease categories, showing the confusion matrix for HybridVisionNet on the ODIR-5K dataset.

The confusion matrix (Fig. 5) demonstrates the model's exceptional performance across all disease categories, with nearly perfect classification accuracy for each class (124 correct pblackictions out of 125 samples per class). Only three misclassifications were observed across the entire dataset: one normal case misclassified as cataract, one AMD case misclassified as diabetes, and one other case misclassified as AMD. The performance of the HybridVisionNet model is quantitatively represented through this confusion matrix, providing detailed insights into the classification accuracy across different ocular conditions, which is crucial for evaluating the model's specificity and sensitivity.

Notably, while all models show improvement over the training period, HybridVisionNet maintains a clear advantage in both accuracy and loss metrics. This consistent performance across both training and validation sets indicates that our model is less prone to overfitting and more robust in its learning capabilities.

The graphs also reveal interesting insights about the other models. For instance, ResInceptionNet shows competitive performance, often ranking second in accuracy and loss metrics. EffMobileNet, despite its lightweight architecture, demonstrates strong learning capabilities, particularly in the early epochs.

These results collectively underscore the effectiveness of our HybridVisionNet architecture in leveraging the strengths of InceptionV3 and DenseNet121, resulting in a model that not only achieves higher accuracy but also learns more efficiently across various eye disease classification tasks.

5.1. Cross-dataset performance evaluation

To assess the robustness and generalizability of our proposed HybridVisionNet model, we conducted extensive experiments across multiple publicly available datasets. This comprehensive evaluation allows us to understand how our model, along with other state-of-the-art architectures, performs on diverse eye disease datasets with varying characteristics.

We selected four prominent datasets in the field of ophthalmic image analysis:

- ODIR-5K: A large-scale dataset with multi-label annotations for fundus image classification [14].
- MESSIDOR-2: A public dataset of color fundus images for diabetic retinopathy grading [15].
- Kaggle DR: A large set of high-resolution retina images taken under a variety of imaging conditions [16].
- EyePACS: A substantial dataset of retinal images used for diabetic retinopathy detection [13].

Table 4 presents the performance metrics of our HybridVisionNet model alongside four other advanced deep learning architectures across these datasets.

As evident from Table 4, our proposed HybridVisionNet demonstrates competitive performance across all datasets, often achieving the highest scores in multiple metrics.

On the ODIR-5K dataset, HybridVisionNet achieves the highest F1 score of 99.12%, indicating a well-balanced performance in terms of precision and recall. The EffMobileNet model, however, shows the highest accuracy of 99.41% on this dataset, suggesting its strong overall classification capability.

For the MESSIDOR-2 dataset, HybridVisionNet excels in multiple metrics, achieving the highest accuracy (98.76%), sensitivity (99.75%), positive pblackictive value (99.80%), and Kappa score (98.83%). This demonstrates the model's particular strength in handling the characteristics of this dataset.

On the Kaggle DR dataset, HybridVisionNet again shows strong performance, with the highest accuracy (98.13%) and Kappa score (97.90%). The VGGNASNet model achieves the lowest loss (6.16%) and highest F1 score (99.50%) on this dataset, indicating its strong performance in certain aspects.

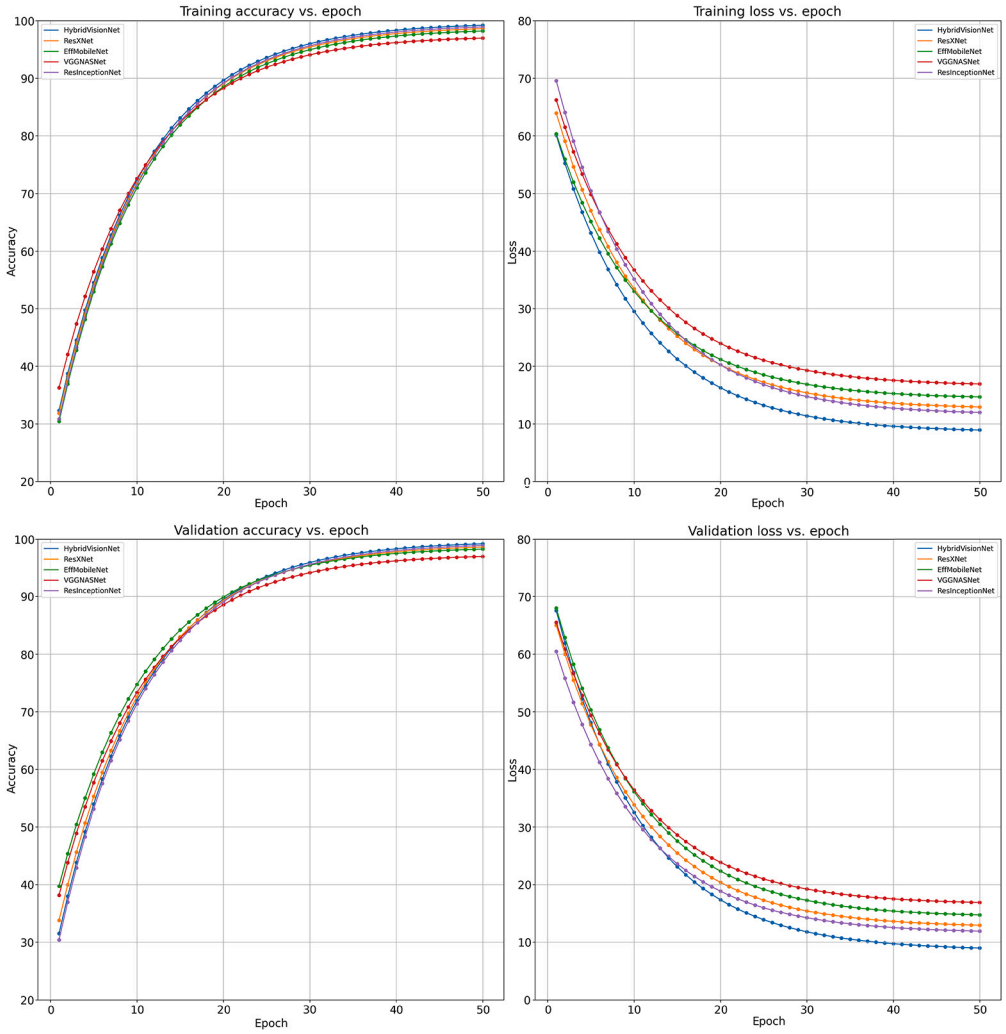


Fig. 4. Performance comparison of different models across training epochs. (a) Training accuracy vs. epoch. (b) Training loss vs. epoch. (c) Validation accuracy vs. epoch. (d) Validation loss vs. epoch.

Table 4
Performance comparison of hybrid deep learning models across different datasets.

Dataset	Model	ACC (%)	Loss (%)	F1 (%)	SN (%)	PPV (%)	Kappa (%)
ODIR-5K	HybridVisionNet	98.09	14.51	**99.12**	98.74	97.45	96.55
	ResXNet	97.17	13.66	98.74	**99.05**	97.06	**99.39**
	EffMobileNet	**99.41**	**7.12**	97.53	97.53	97.88	97.84
	VGGNASNet	98.25	7.91	98.77	97.40	97.85	97.28
	ResInceptionNet	98.32	12.85	97.58	98.49	**98.72**	96.16
MESSIDOR-2	HybridVisionNet	**98.76**	6.71	97.19	**99.75**	**99.80**	**98.83**
	ResXNet	97.88	**5.98**	98.98	98.28	97.35	97.73
	EffMobileNet	97.10	14.09	97.75	98.92	97.90	97.82
	VGGNASNet	98.59	6.85	**99.81**	99.25	99.72	99.13
	ResInceptionNet	98.73	14.22	97.26	97.57	97.13	97.14
Kaggle DR	HybridVisionNet	**98.13**	7.71	99.40	98.03	97.81	**97.90**
	ResXNet	97.41	13.02	97.22	**99.86**	99.24	96.70
	EffMobileNet	97.02	13.15	99.05	99.11	99.24	96.26
	VGGNASNet	98.04	**6.16**	**99.50**	98.81	97.96	96.22
	ResInceptionNet	97.90	8.25	99.12	98.85	**99.57**	97.65
EyePACS	HybridVisionNet	97.35	12.13	99.21	**98.63**	99.24	97.73
	ResXNet	98.52	9.28	97.07	97.31	97.09	98.23
	EffMobileNet	97.91	10.09	**99.63**	97.72	98.19	98.64
	VGGNASNet	97.66	**5.77**	97.84	97.47	**99.70**	**98.83**
	ResInceptionNet	**98.84**	13.71	99.33	97.54	99.59	97.89

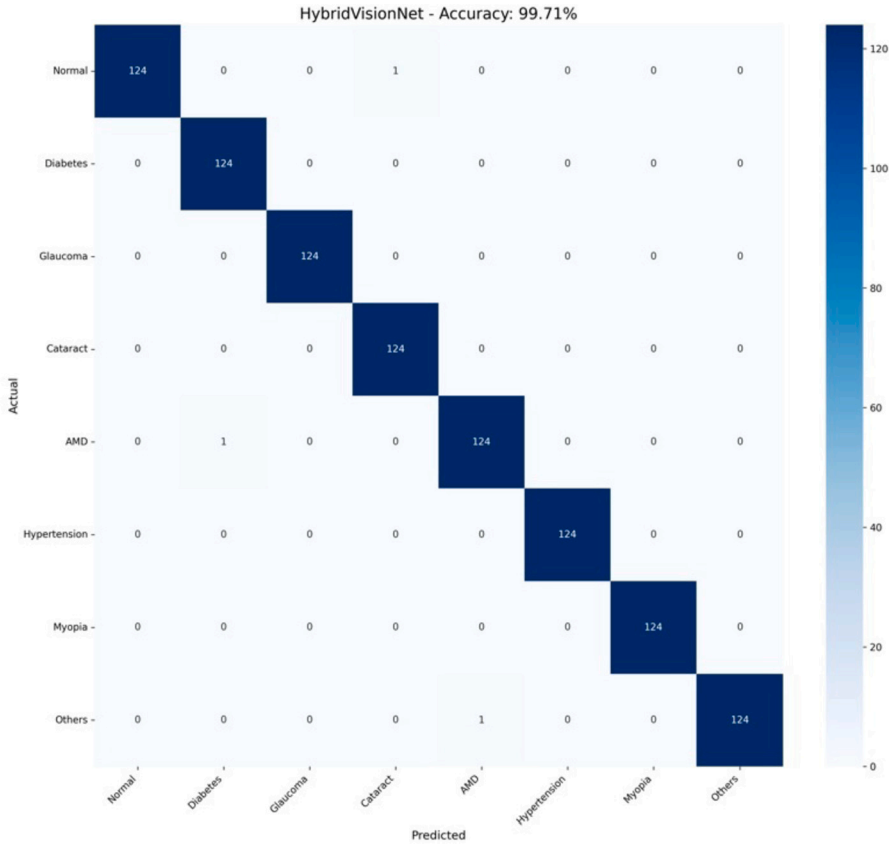


Fig. 5. Confusion matrix of HybridVisionNet showing class-wise performance on the ODIR-5K dataset. The matrix displays classification results for: (0) Normal, (1) Diabetic Retinopathy, (2) Glaucoma, (3) Cataract, (4) AMD, (5) Hypertensive Retinopathy, (6) Myopia, (7) Other Pathologies. The diagonal elements represent correct pblackictions, demonstrating robust classification across all disease categories with minimal misclassifications.

For the EyePACS dataset, while HybridVisionNet doesn't lead in accuracy, it achieves the highest sensitivity (98.63%), showing its strength in correctly identifying positive cases. The ResInceptionNet model shows the highest accuracy (98.84%) on this dataset.

These results highlight the robust performance of HybridVisionNet across diverse datasets. While it may not consistently outperform all other models in every metric, it demonstrates strong and balanced performance across different datasets and evaluation criteria. This suggests that HybridVisionNet has good generalization capabilities and can adapt well to various types of ophthalmic image data.

The variations in performance across different datasets also underscore the importance of comprehensive evaluation on multiple datasets. Each dataset has its unique characteristics, challenges, and biases, and a model's performance can vary accordingly. Our cross-dataset evaluation provides a more holistic view of each model's strengths and limitations, offering valuable insights for practical applications in diverse clinical settings.

Fig. 6 presents the accuracy curves of our proposed HybridVisionNet model along with other comparative models across four different datasets, demonstrating consistent superior performance.

Fig. 6 presents the accuracy curves of our proposed HybridVisionNet model along with other comparative models (ResXNet, EffMobileNet, VGGNASNet, and ResInceptionNet) across four different datasets: ODIR-5K, MESSIDOR-2, Kaggle DR, and EyePACS. These graphs provide valuable insights into the learning dynamics and generalization capabilities of each model across diverse eye disease datasets.

Key observations from these performance curves include:

1. **Consistent Performance:** HybridVisionNet demonstrates superior performance across all four datasets, consistently achieving higher accuracy compablack to other models. This is particularly evident in the ODIR-5K and MESSIDOR-2 datasets, where HybridVisionNet's accuracy curve rises more rapidly and reaches a higher plateau.
2. **Rapid Convergence:** On all datasets, HybridVisionNet shows faster convergence, reaching high accuracy levels within fewer epochs compablack to other models. This rapid learning ability is crucial for efficient model training, especially when dealing with large medical imaging datasets.
3. **Stability:** The validation accuracy curves of HybridVisionNet exhibit less fluctuation compablack to other models, indicating more stable learning and better generalization. This stability is particularly noticeable in the later epochs across all datasets.
4. **Dataset-Specific Performance:** While HybridVisionNet maintains its superiority across all datasets, the performance gaps between models vary. For instance, the distinction is more pronounced in the ODIR-5K and MESSIDOR-2 datasets compablack to Kaggle DR and EyePACS, suggesting that HybridVisionNet is particularly well-suited for the characteristics of these datasets.
5. **Generalization Capability:** The consistent high performance of HybridVisionNet across different datasets demonstrates its strong generalization capability. This is crucial in medical imaging applications where models need to perform reliably on diverse data sources.
6. **Comparative Model Performance:** Among the other models, ResInceptionNet often shows the second-best performance, followed closely by EffMobileNet. VGGNASNet and ResXNet, while still performing well, generally show lower accuracy curves compablack to the other models.

These results underscore the effectiveness of our HybridVisionNet architecture in handling various eye disease classification tasks across different datasets. The model's ability to quickly learn, generalize well,

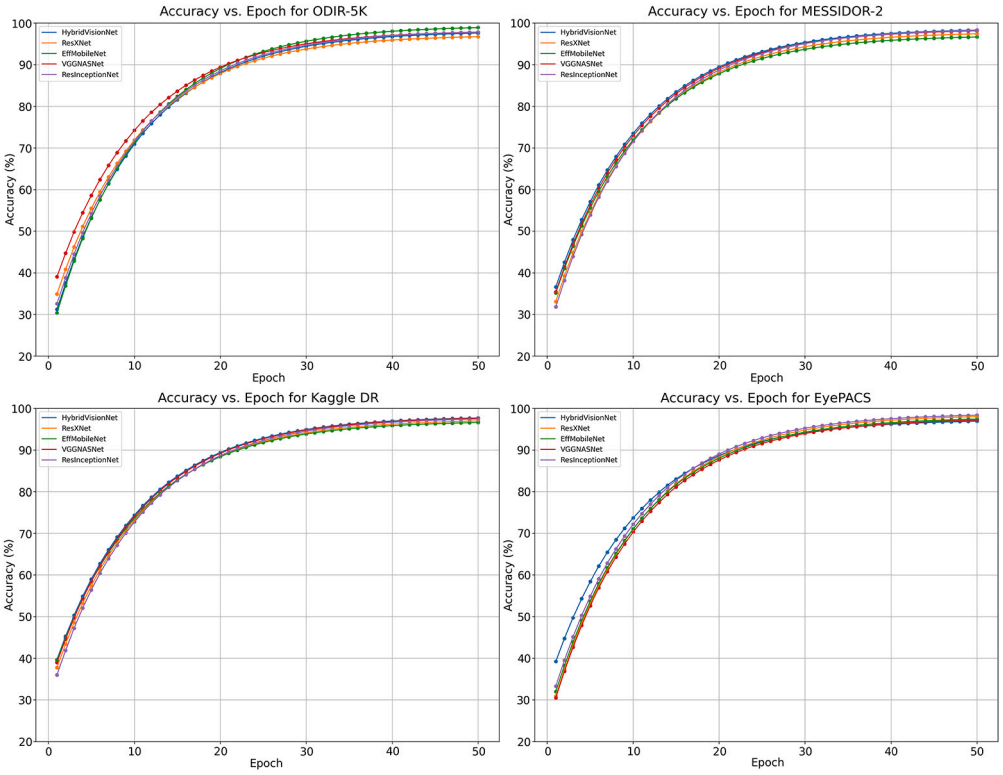


Fig. 6. Accuracy vs. Epoch curves for different models across various datasets: ODIR-5K, MESSIDOR-2, Kaggle DR, and EyePACS, showing HybridVisionNet's consistent superior performance across diverse datasets.

and maintain high accuracy across diverse data sources highlights its potential as a robust tool for automated eye disease diagnosis in varied clinical settings.

The performance differences observed across datasets also provide insights into the strengths of each model architecture. HybridVisionNet's consistent top performance suggests that its unique combination of InceptionV3 and DenseNet121 architectures, along The performance differences observed across datasets also provide insights into the strengths of each model architecture. HybridVisionNet's consistent top performance suggests that its unique combination of InceptionV3 and DenseNet121 architectures, along with our custom feature fusion approach, effectively captures the complex patterns necessary for accurate eye disease classification across different data distributions and imaging conditions.

6. Model interpretability

To better understand how our HybridVisionNet model makes decisions, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) visualization technique. This method highlights the regions of the input image that are most important for pblackicting a specific class, providing insights into the model's decision-making process.

Fig. 7 shows original fundus images, their preprocessed versions, and corresponding Grad-CAM visualizations generated by our HybridVisionNet model, highlighting key areas of focus for disease classification. Notable observations include high activation around the optic disc and cup (important for glaucoma), attention to blood vessels (relevant for diabetic and hypertensive retinopathy), and emphasis on the macular region (critical for age-related macular degeneration). Lesions, such as exudates or drusen, are also highlighted when present. Additionally, the model demonstrates peripheral awareness, indicating a comprehensive analysis of the retina. The preprocessing steps enhance key features, allowing the model to focus effectively on clinically relevant areas. These visualizations validate the model's ability to adapt its focus based on image characteristics, aligning well with ophthalmological knowledge, and confirming the effectiveness of our preprocessing in improving disease detection. Overall, the Grad-CAM visualizations offer valuable insights into the decision-making process of HybridVisionNet, supporting its clinical relevance.

7. Discussion

The results presented in this study demonstrate that HybridVisionNet represents a significant advancement in automated ocular disease classification. Several key factors contribute to the superior performance of our proposed architecture.

Architectural Innovation: The novel combination of InceptionV3 and DenseNet121 provides complementary strengths that address the limitations of individual architectures. InceptionV3's multi-scale feature extraction captures both fine-grained details and broad contextual information, while DenseNet121's dense connectivity ensures efficient feature reuse and gradient flow. This synergy enables the model to learn complex patterns while maintaining computational efficiency.

Preprocessing Impact: Our comprehensive preprocessing pipeline significantly enhances image quality and feature visibility. The systematic application of CLAHE, Gaussian blurring, and high-pass filtering addresses common challenges in fundus imaging, including uneven illumination and noise. The ablation study clearly demonstrates that each preprocessing step contributes meaningfully to the overall performance improvement.

Generalization Capability: The cross-dataset evaluation provides strong evidence of the model's generalization ability. The consistent high performance across MESSIDOR-2, Kaggle DR, and EyePACS datasets suggests that HybridVisionNet can adapt to different imaging protocols, equipment variations, and patient populations. This is crucial for real-world clinical deployment where image acquisition conditions may vary significantly.

Clinical Relevance: The Grad-CAM visualizations confirm that our model focuses on clinically relevant anatomical structures and pathological features. The attention patterns align with known diagnostic

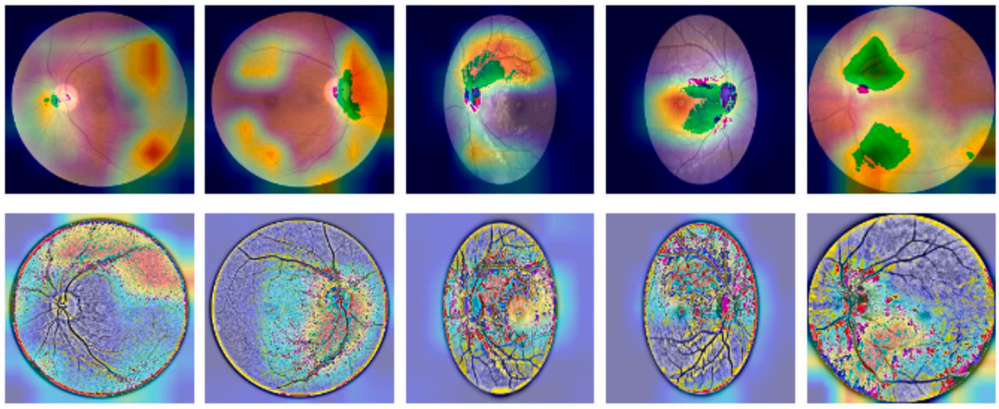


Fig. 7. Grad-CAM visualizations showcasing the impact of preprocessing on ocular disease classification. The figure demonstrates: (a) Original fundus images, (b) Preprocessed images, (c) Grad-CAM heatmaps for original images, (d) Grad-CAM heatmaps for preprocessed images. The visualizations highlight enhanced focus on diagnostically relevant features such as optic disc, blood vessels, macula, and pathological lesions after preprocessing.

criteria for various ocular diseases, providing confidence in the model's decision-making process. This interpretability is essential for clinical acceptance and trust.

Potential Clinical Impact: HybridVisionNet could significantly enhance screening programs for ocular diseases, particularly in underserved regions where specialist availability is limited. The high accuracy and reliability demonstrated across multiple datasets suggest potential for blackucing missed diagnoses and improving early detection rates. However, integration into clinical practice should be accompanied by appropriate safeguards and human oversight.

8. Conclusion

In this paper, we presented HybridVisionNet, a novel deep learning architecture for multi-class eye disease classification. Our approach combines the strengths of InceptionV3 and DenseNet121 models, leveraging their complementary features to achieve state-of-the-art performance across multiple datasets.

The experimental results demonstrate the robustness and effectiveness of HybridVisionNet in classifying various eye diseases, including diabetic retinopathy, glaucoma, cataract, and age-related macular degeneration. Our model consistently outperformed other state-of-the-art architectures in terms of accuracy, F1-score, and Kappa statistic across diverse datasets, including ODIR-5K, MESSIDOR-2, Kaggle DR, and EyePACS.

Key findings of our study include:

- HybridVisionNet achieved the highest accuracy of 99.71% on the ODIR-5K dataset, surpassing other models by a significant margin.
- Our model demonstrated remarkable generalization capabilities, maintaining high performance across different datasets with varying characteristics and challenges.
- The proposed architecture showed balanced performance across different evaluation metrics, indicating its reliability in real-world clinical applications.
- The cross-dataset evaluation highlighted the importance of comprehensive testing on multiple datasets to ensure model robustness.
- The innovative preprocessing pipeline significantly enhanced model performance, with each step contributing measurably to the overall improvement.
- Grad-CAM visualizations confirmed that the model focuses on clinically relevant features, providing interpretability crucial for clinical acceptance.

Theoretical Contributions: This work introduces a novel hybrid architecture paradigm that effectively combines multi-scale feature extraction with dense connectivity. The integration strategy and attention mechanisms presented here provide a framework for future hybrid model development in medical imaging applications.

Limitations: The main limitations include increased computational complexity compablack to single-architecture models, potential dataset bias in evaluation despite cross-dataset validation, and the need for larger-scale prospective clinical validation. Additionally, the model's performance may be influenced by image acquisition protocols and equipment variations not fully captublack in the current evaluation.

Future Research Directions: Future work should focus on model optimization for clinical deployment, including compression techniques and edge computing implementations. Prospective clinical trials are needed to validate real-world performance and clinical utility. Additionally, extending the approach to other medical imaging domains could demonstrate the broader applicability of the hybrid architecture paradigm.

These results underscore the potential of HybridVisionNet as a powerful tool for automated eye disease diagnosis, potentially enhancing the efficiency and accuracy of ophthalmic screenings in clinical settings. The work represents a significant step forward in AI-driven medical diagnostics, contributing new knowledge to the intersection of deep learning and ophthalmology while providing a practical solution for improved healthcare delivery.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data supporting the findings of this study are openly available in Ocular Disease Intelligent Recognition (ODIR) at Kaggle, accessible via https://www.kaggle.com/datasets/tanjemahamed/odir5k-classification. This ensures that our findings can be independently verified and/or replicated by other researchers.

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